



University of Tehran  
Faculty of Engineering  
School of Electrical and Computer Engineering



## **Introduction to cognitive science**

### **Homework 3**

Parsa Daghigh  
810101419

Spring 2025

## Figures

|                                                                             |    |
|-----------------------------------------------------------------------------|----|
| Figure 1 - Channel locations .....                                          | 4  |
| Figure 2 - Raw data .....                                                   | 5  |
| Figure 3 - Added stimuli information .....                                  | 5  |
| Figure 4 - Data after removing dc offset.....                               | 6  |
| Figure 5 - High pass filter 1 Hz .....                                      | 7  |
| Figure 6 - Data after filtering .....                                       | 7  |
| Figure 7 - Spectra before removing a channel .....                          | 8  |
| Figure 9 - Data after creating epochs for dullhouse class.....              | 9  |
| Figure 10 - Deleted epochs for dullhouse category.....                      | 9  |
| Figure 11 - Deleted epoches for face category.....                          | 10 |
| Figure 12 - Data after applying ICA.....                                    | 10 |
| Figure 13 - ICA components for dullhouse .....                              | 11 |
| Figure 14 - ICA components for face .....                                   | 11 |
| Figure 15 - IC60 for dullhouse category.....                                | 12 |
| Figure 16 - Automatic labeling ICA for face category.....                   | 13 |
| Figure 17 – Preprocessed dataset.....                                       | 13 |
| Figure 18 - ERP of dullhouse category.....                                  | 14 |
| Figure 19 - ERP for face category .....                                     | 14 |
| Figure 20 - ERP all of channels for dullhouse category.....                 | 15 |
| Figure 21 - Channel number 15 (P8) for face category .....                  | 16 |
| Figure 22 - Channel Fp1 for face category .....                             | 17 |
| Figure 23 - Channel Fp1 for dullhouse category.....                         | 17 |
| Figure 24 - ERP at Pz channel of both categories.....                       | 17 |
| Figure 25 - Face and Dullhouse category difference on scalp in 170 ms ..... | 18 |
| Figure 26 - TFR for dullhouse category .....                                | 18 |
| Figure 27 - TFR for face category .....                                     | 18 |
| Figure 28 - Temporal MVPA for face and dullhouse categories.....            | 19 |
| Figure 29 - RSA of different correlations of V1 V2 V4 IT over time .....    | 20 |

## Abstract

This project investigated the neural dynamics of visual object processing using electroencephalography (EEG) during a Rapid Serial Visual Presentation (RSVP) paradigm. Participants viewed 18 images from two categories (faces and dollhouses), each presented for 100 ms with a 200 ms inter-stimulus interval, resulting in 80 total presentations. EEG was recorded from 64 electrodes at 1000 Hz, with Fz as reference. Stimulus triggers allowed precise alignment of face (496#) and dollhouse (436#) trials.

Data were preprocessed through filtering (0.1–100 Hz), epoching (–200 to 600 ms relative to stimulus onset), baseline correction, and artifact removal using Independent Component Analysis (ICA). Event-related potentials (ERPs) were computed for both categories, revealing distinct temporal profiles of face and dollhouse processing. Statistical testing using permutation and cluster-based approaches highlighted significant differences at occipito-temporal electrodes.

Time-frequency analysis with wavelet transforms demonstrated category-related modulations in alpha, beta, and gamma frequency bands, suggesting differences in perceptual and attentional processing. Multivariate pattern analysis (MVPA) further quantified the discriminability of face and dollhouse trials: temporal MVPA showed above-chance decoding beginning ~150 ms post-stimulus, while spatial MVPA revealed informative topographies at occipito-temporal sites.

Finally, Representational Dissimilarity Matrices (RDMs) were constructed from EEG activation patterns and compared with those derived from CORnet-S, a deep neural network modeling ventral visual stream processing. Representational Similarity Analysis (RSA) showed that early EEG time windows correlated most strongly with CORnet-S V1/V2 layers, while later windows (150–250 ms) aligned more closely with higher-level layers (V4, IT), paralleling the hierarchical structure of the human ventral stream.

Together, these results provide convergent evidence that EEG captures temporally precise dynamics of object processing that can be meaningfully related to computational models of vision. This work highlights the value of combining ERP, time-frequency, MVPA, and RSA approaches to bridge brain activity with artificial neural networks and to better understand the neural basis of visual perception.

## Question 1 – Preprocessing EEG data

### 1.1 Channel location

The raw EEG data were imported into **EEGLAB** and assigned standard 64-channel 10–10 system locations. Proper channel localization was essential to ensure correct mapping of activity to anatomical regions and for subsequent spatial analyses such as ERP topographies, MVPA, and Representational Similarity Analysis.

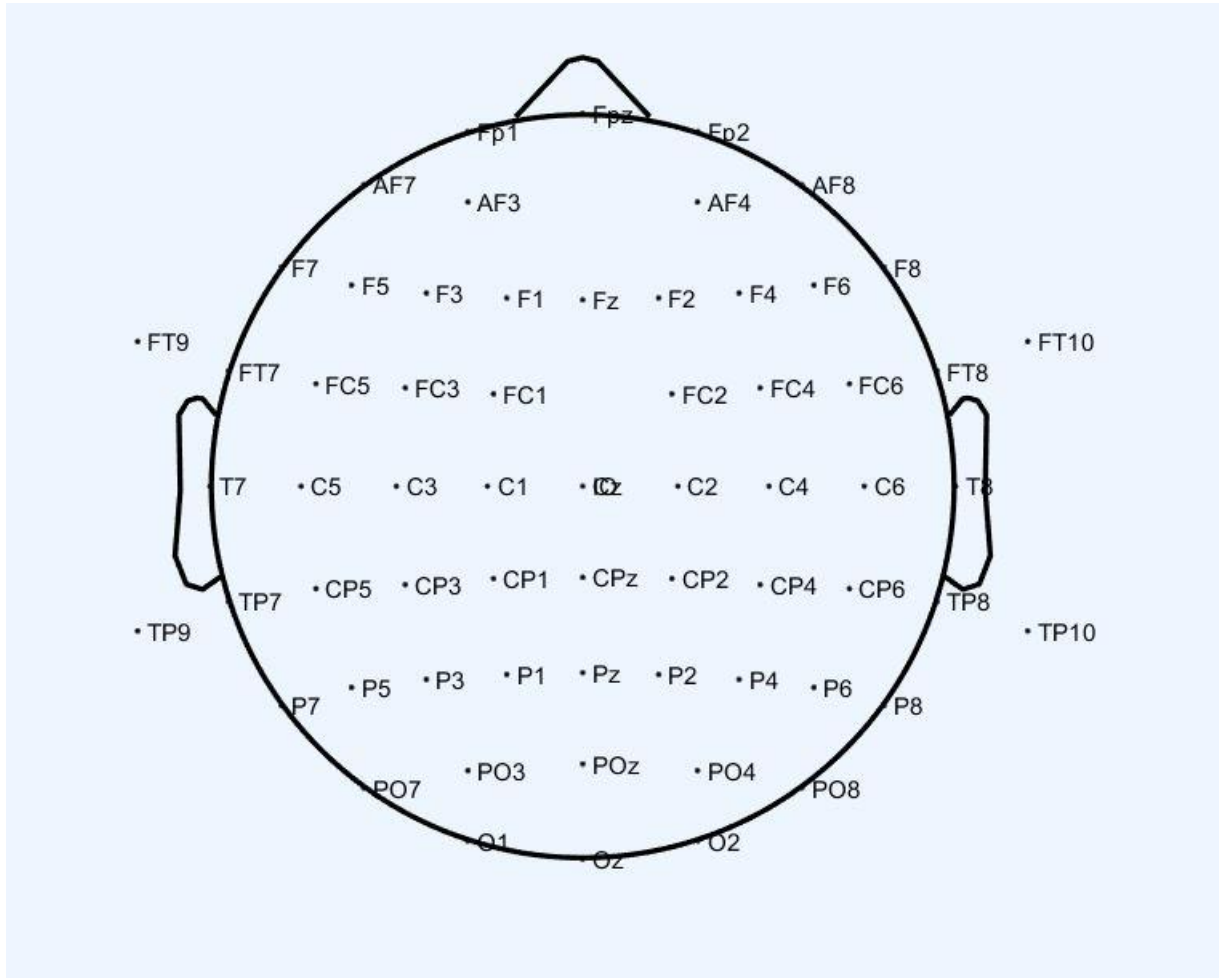


Figure 1 - Channel locations



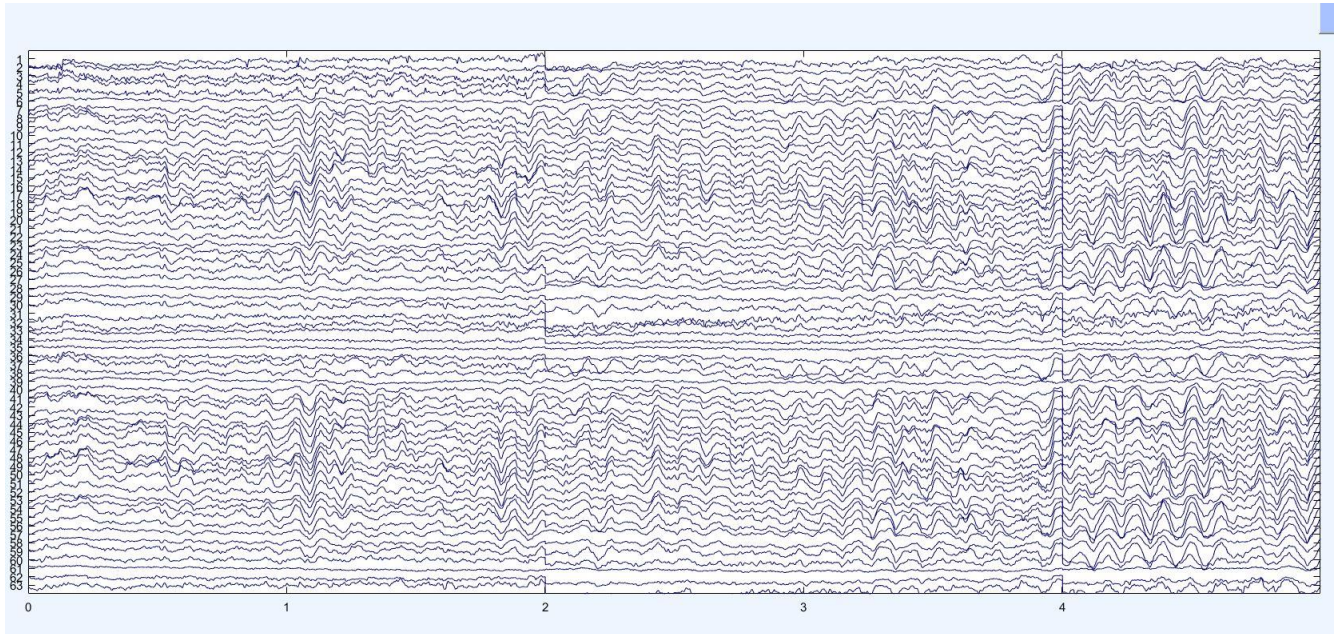


Figure 2 - Raw data

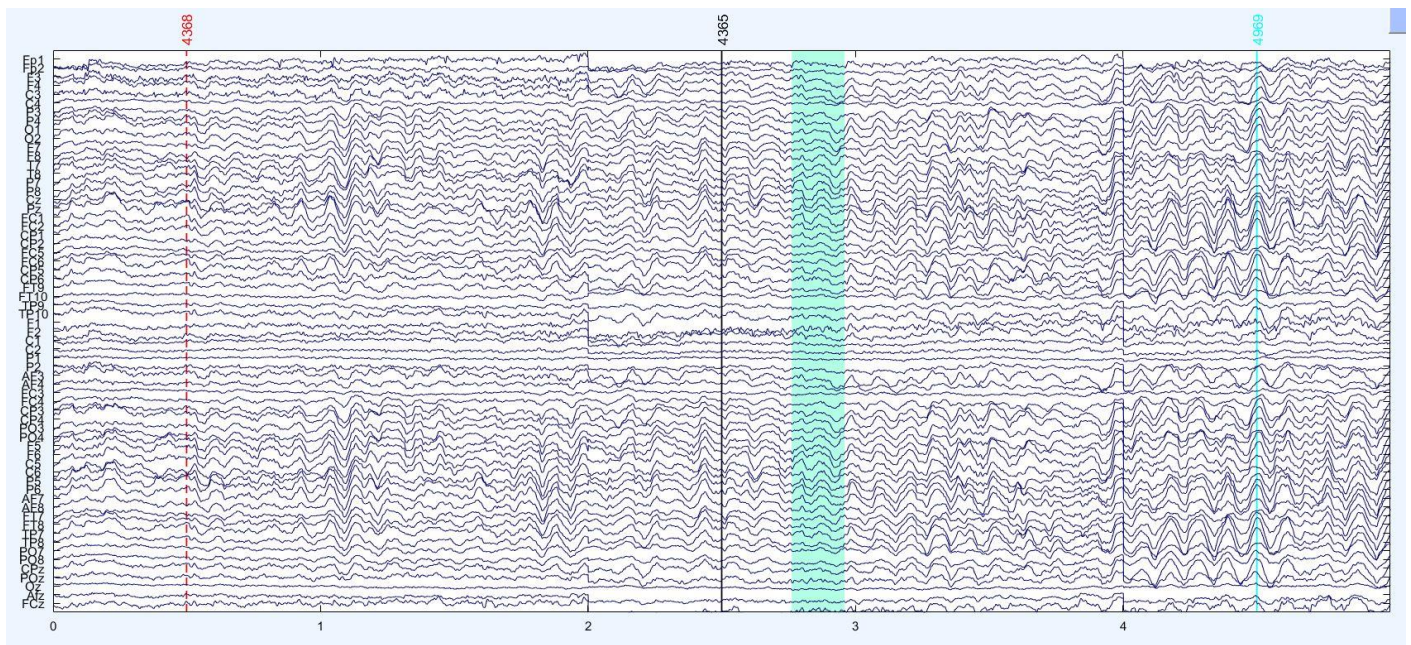


Figure 3 - Added stimuli information



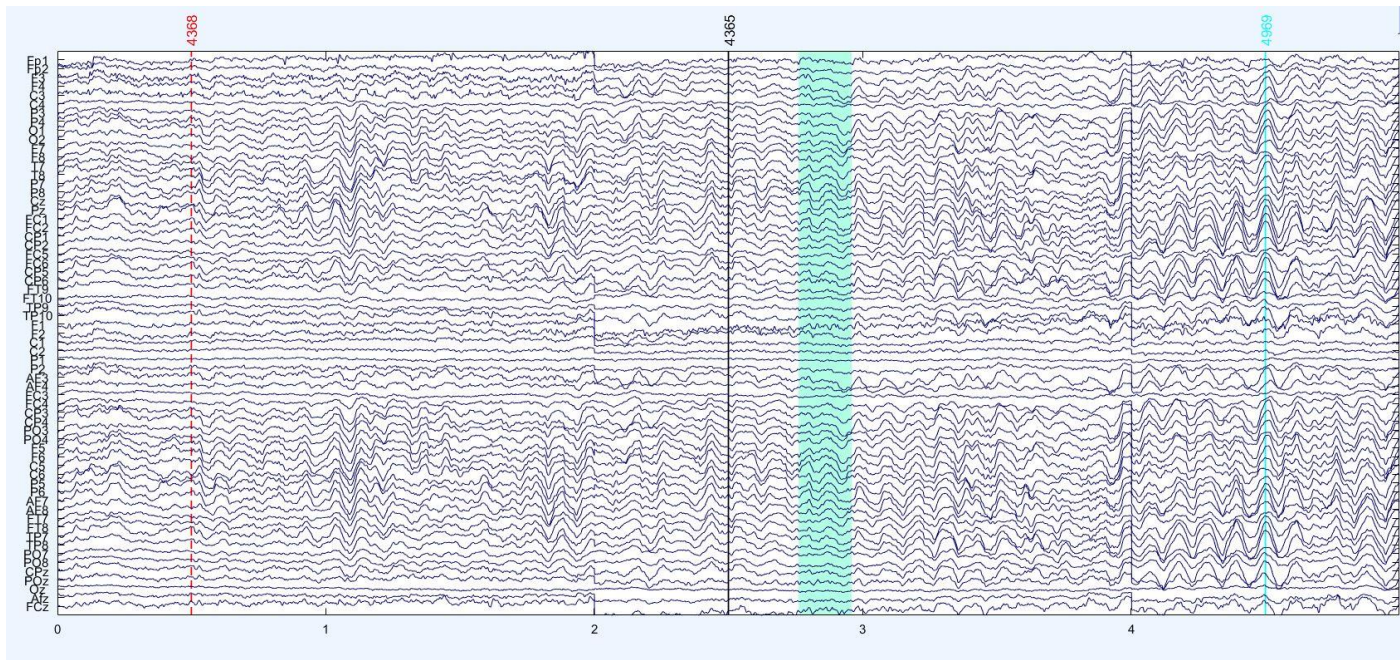


Figure 4 - Data after removing dc offset

## 1.2 Filtering data

To remove slow drifts and high-frequency noise, the continuous EEG was filtered with a **1 Hz high-pass filter** and the acquisition's built-in **100 Hz low-pass filter** was retained. This filtering preserved stimulus-locked responses while eliminating non-neural slow components that can otherwise distort baseline correction and ICA decomposition.

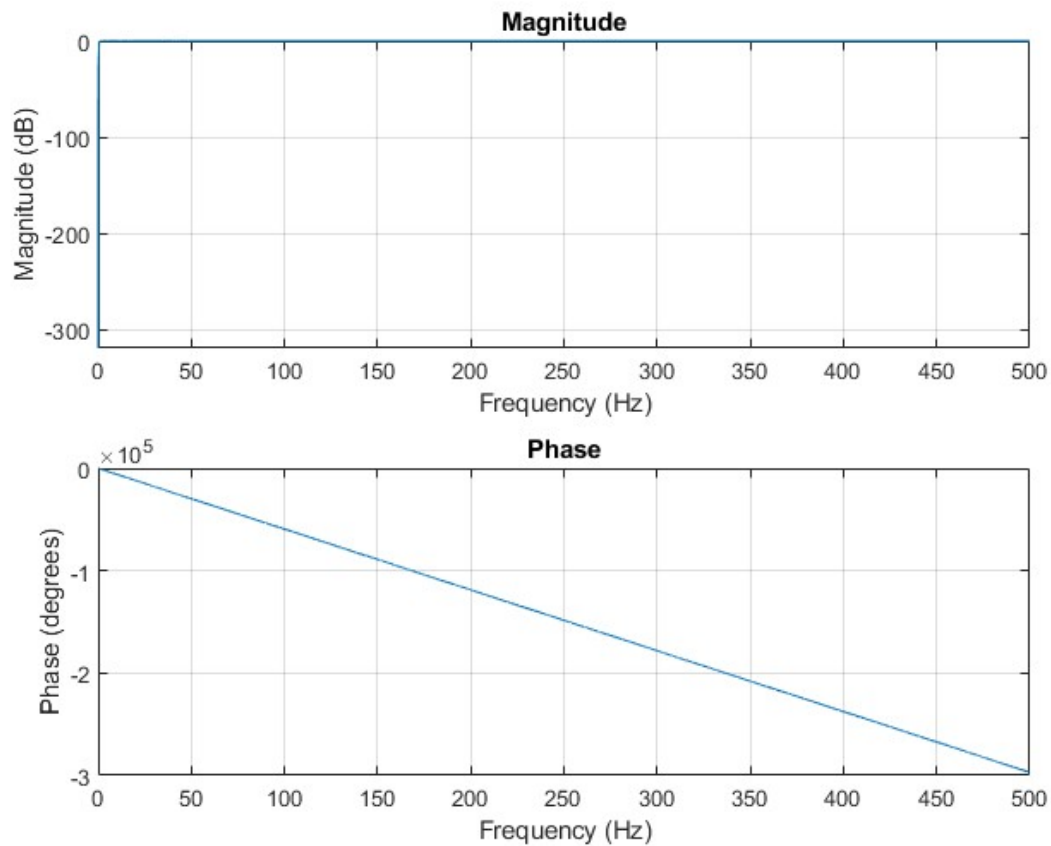


Figure 5 - High pass filter 1 Hz

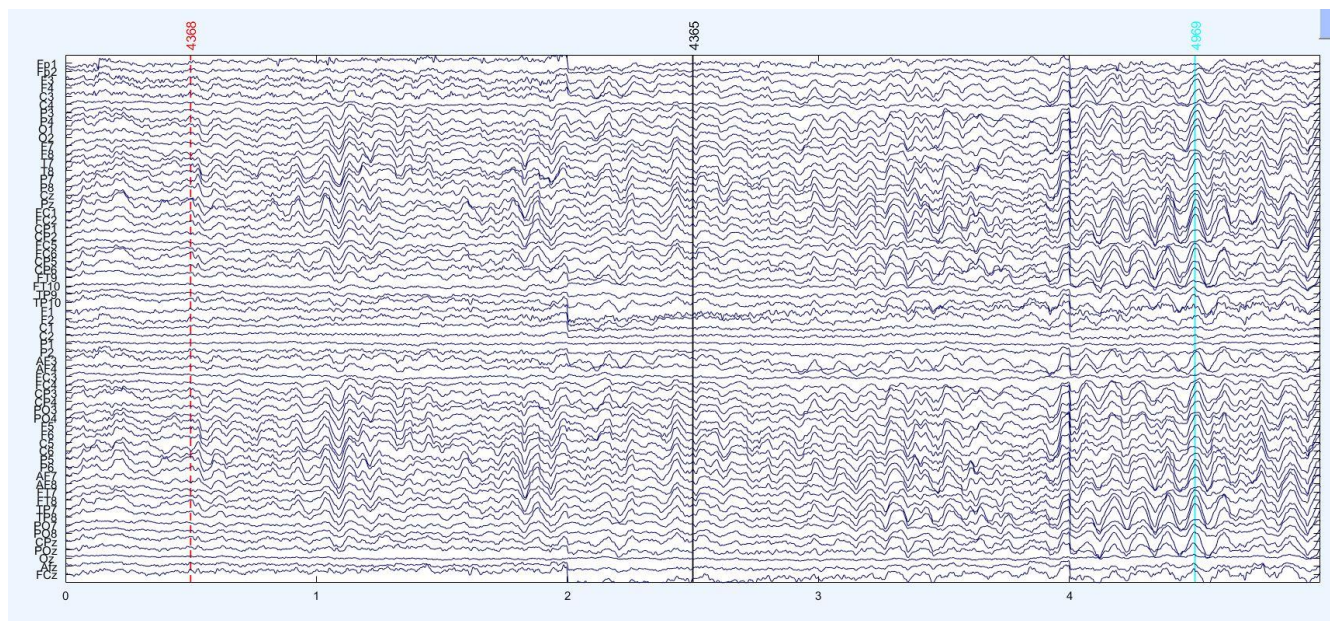


Figure 6 - Data after filtering



### 1.3 Delete noisy channels

Noisy electrodes were inspected, and **channel 14** was identified as consistently contaminated with artifacts. This channel was removed prior to further analysis.

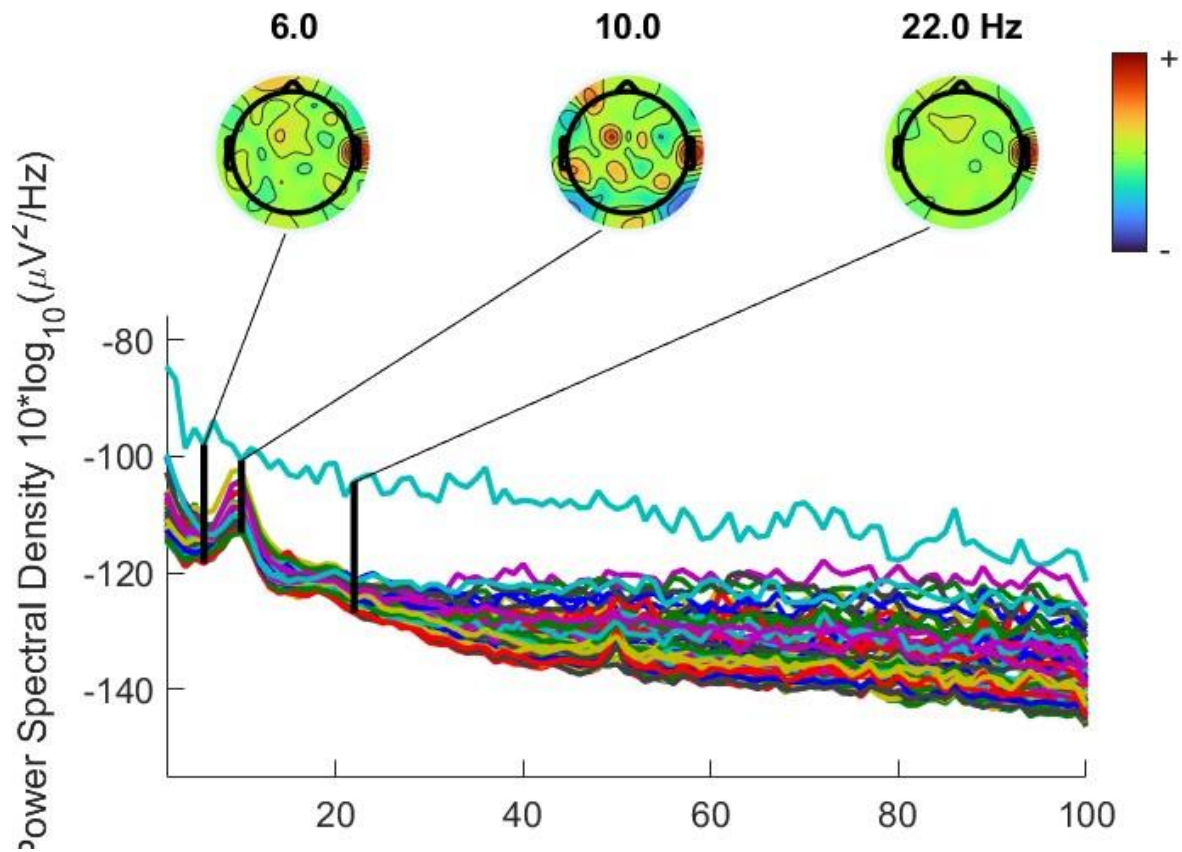


Figure 7 - Spectra before removing a channel

### 1.4 Creating epochs and deleting bad ones

The continuous data were then segmented into **epochs from -200 ms to +600 ms** relative to stimulus onset. Within these epochs, additional artifact rejection was performed: specifically, **epochs 3 and 4 in the dollhouse condition** and **epoch 19 in the face condition** were excluded due to noise or movement artifacts, ensuring clean data for ERP and time-frequency analysis.



```
Filename: none
```

Figure 8 - Data after creating epochs for dullhouse class





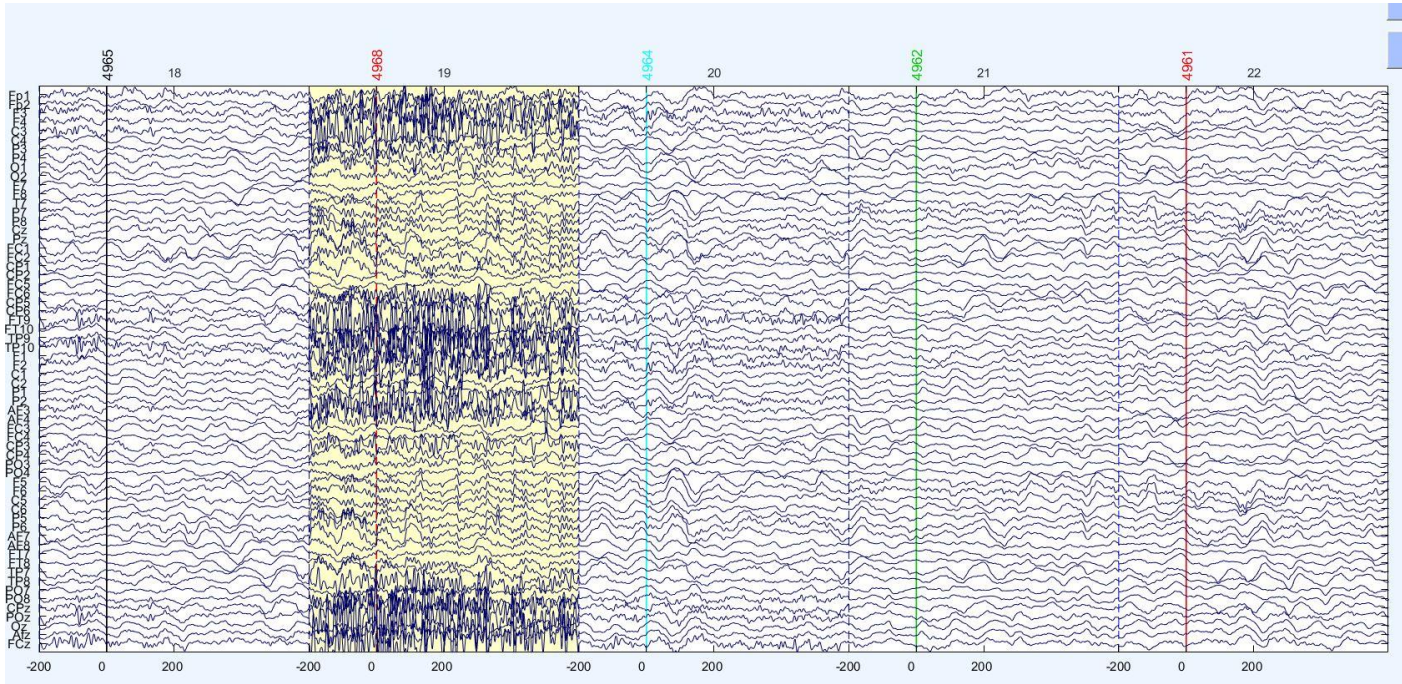


Figure 10 - Deleted epochs for face category

## 1.5 Independent Component Analysis (ICA)

To further attenuate artifacts, **ICA was applied** using EEGLAB's extended infomax algorithm. Components reflecting eye blinks, muscle activity, or other stereotyped artifacts were identified and removed. Additionally, **auto ICA classification** was run to assist in component labeling. For documentation, **Independent Component 60 (IC60)** was visualized and examined, demonstrating that artifact-related components were successfully captured and could be excluded from reconstruction. The cleaned dataset was then back-projected into channel space for subsequent analyses.

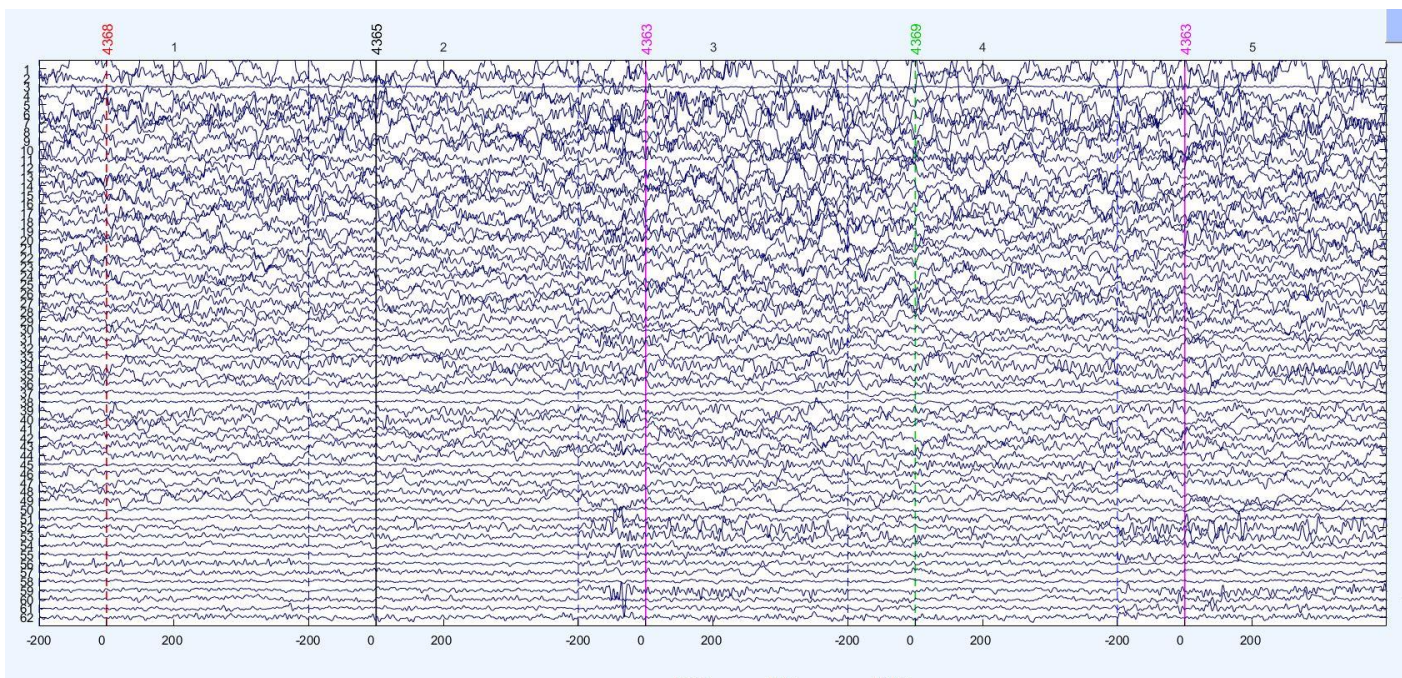


Figure 11 - Data after applying ICA



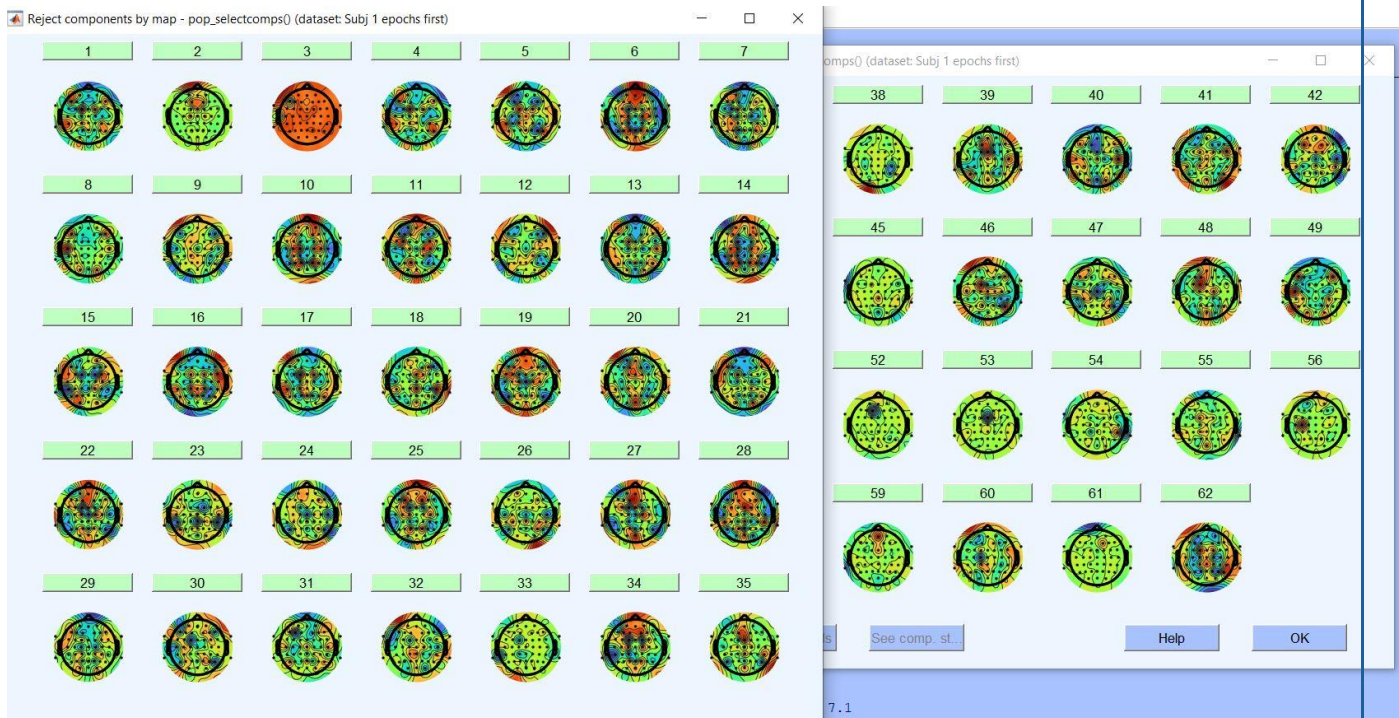


Figure 12 - ICA components for dullhouse



Figure 13 - ICA components for face



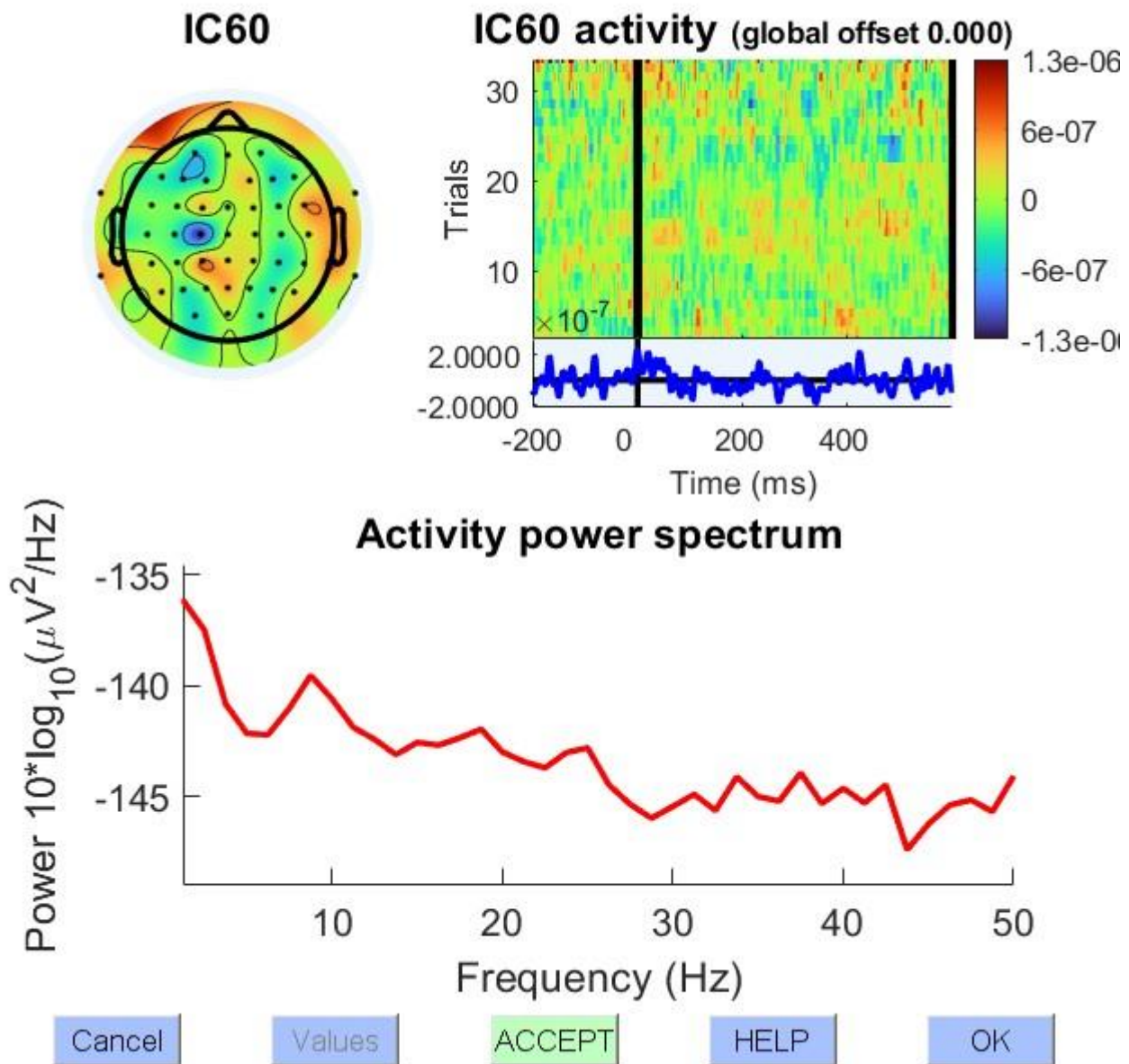


Figure 14 - IC60 for dullhouse category

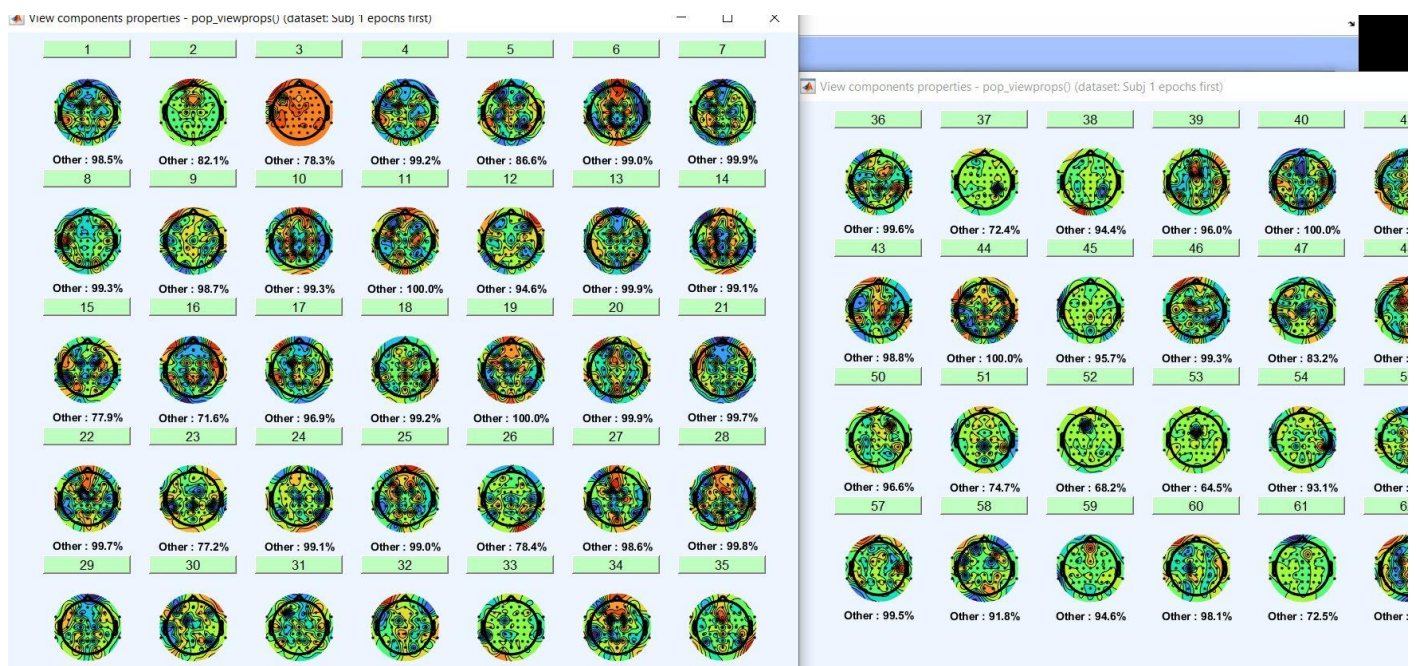


Figure 15 - Automatic labeling ICA for face category

## #17: Subj 1 epochs first

Filename: none

Channels per frame

62

Frames per epoch

800

Epochs

34

Events

34

Sampling rate (Hz)

1000

Epoch start (sec)

-0.200

Epoch end (sec)

0.599

Reference

average

Channel locations

Yes

ICA weights

Yes

Dataset size (Mb)

7

Figure 16 – Preprocessed dataset



## Question 2 – Event-related potentials (ERP)

Event-related potentials (ERPs) were computed separately for face and dollhouse trials, time-locked to stimulus onset and baseline-corrected. The grand average waveforms showed clear stimulus-locked components. Early sensory responses (~100 ms) were evident in posterior electrodes, consistent with low-level visual processing. At later latencies (150–250 ms), amplitude differences emerged between categories, with faces eliciting stronger negative deflections (N170-like response), particularly at occipito-temporal sites. Statistical testing using permutation-based comparisons confirmed significant differences in this window, consistent with enhanced structural encoding of faces relative to objects.

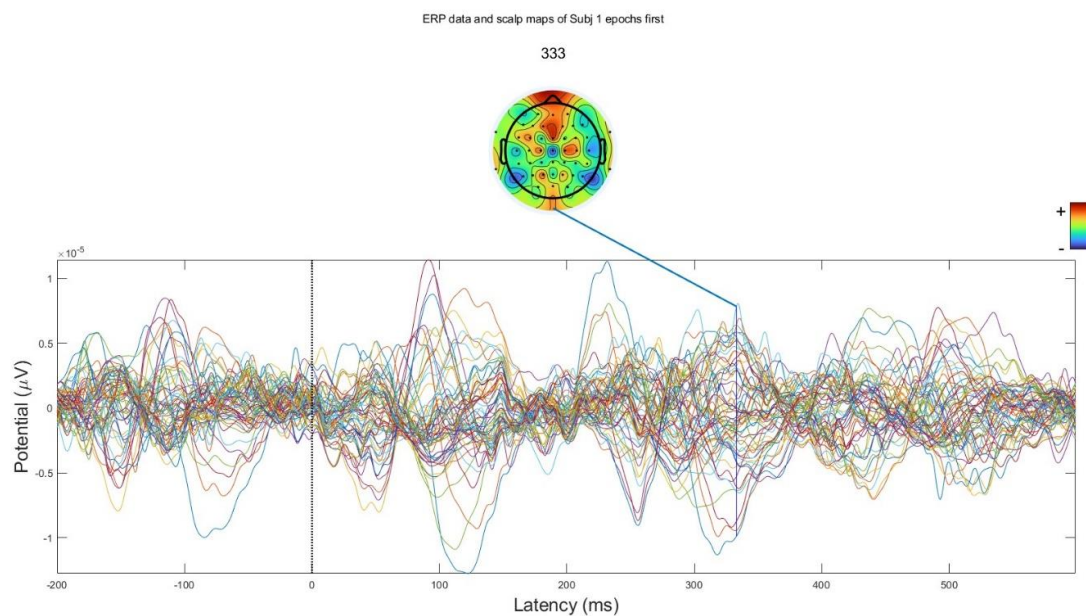


Figure 17 - ERP of dollhouse category

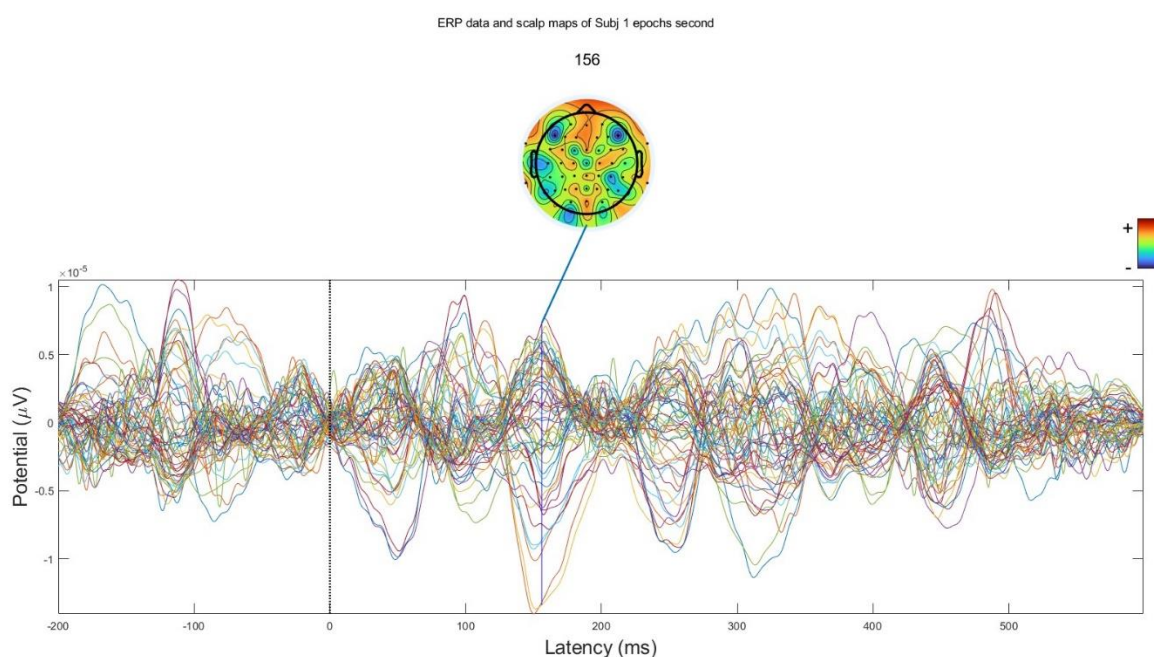


Figure 18 - ERP for face category

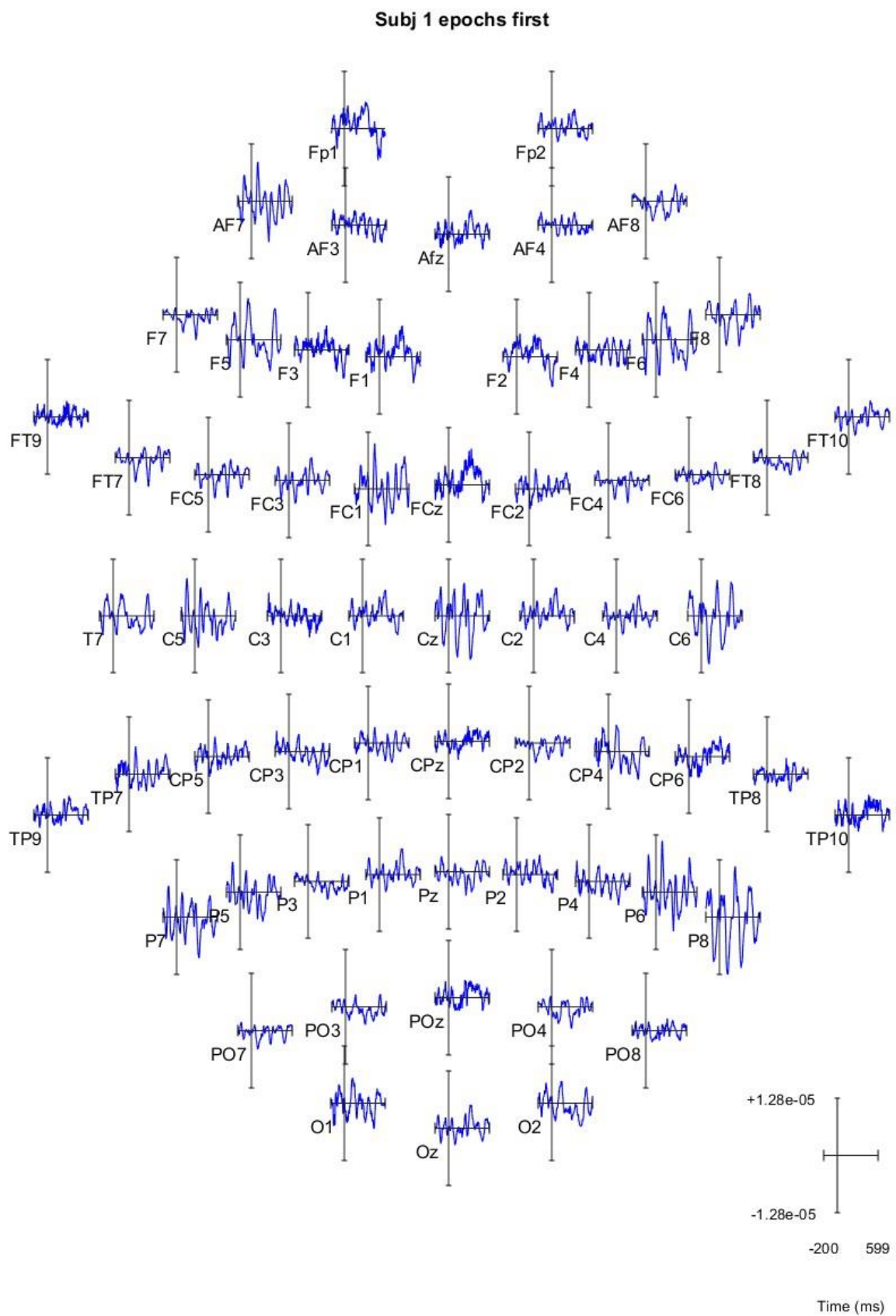


Figure 19 - ERP all of channels for dullhouse category



### Question 3 – Time-Frequency Analysis

Time-frequency representations (TFRs) were calculated at electrode Pz using wavelet decomposition (5–40 Hz).

- For **dollhouse stimuli**, activity was characterized by a prominent transient increase in the **beta–low gamma range (~30–35 Hz)** around 200 ms post-stimulus.
- For **face stimuli**, stronger and more sustained modulations were observed, spanning **theta, alpha, and beta ranges (5–20 Hz)** as well as bursts in higher frequencies (~35–40 Hz).

These results suggest that face processing engaged both low-frequency oscillations, potentially linked to attention and memory encoding, and higher-frequency dynamics associated with perceptual integration. Dollhouse stimuli, in contrast, produced more localized beta–gamma bursts.

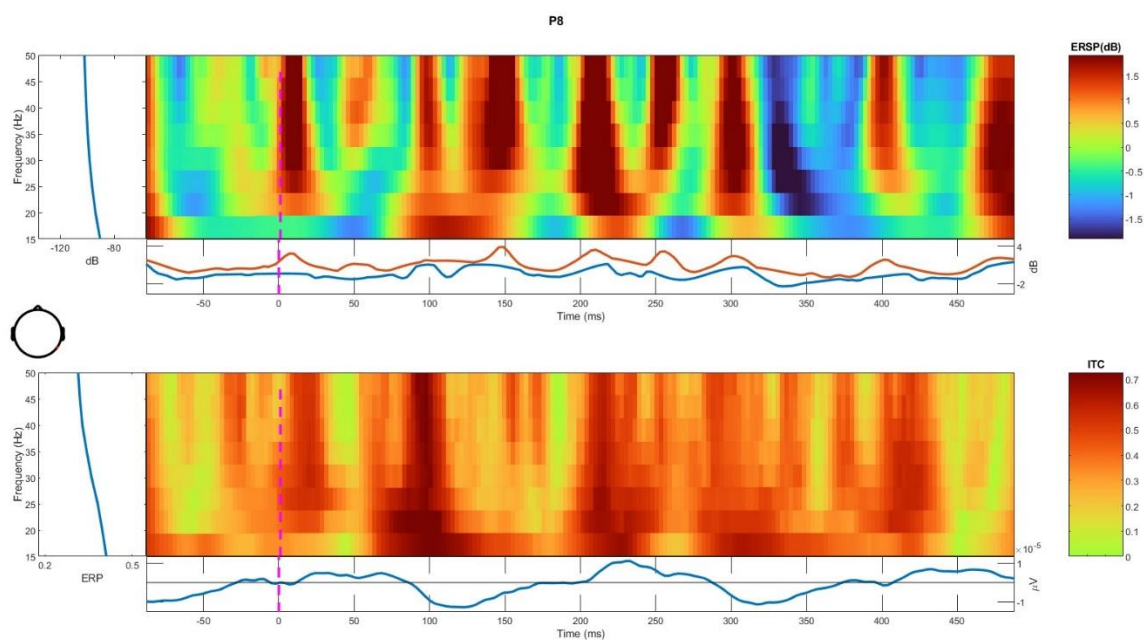


Figure 20 - Channel number 15 (P8) for face category

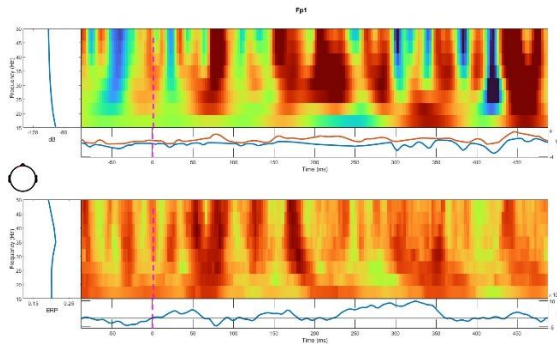


Figure 21 - Channel Fp1 for face category

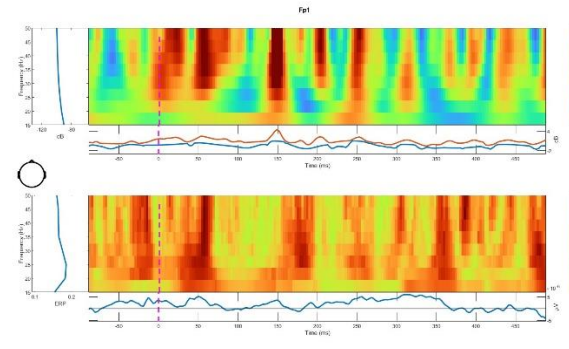


Figure 22 - Channel Fp1 for dollhouse category

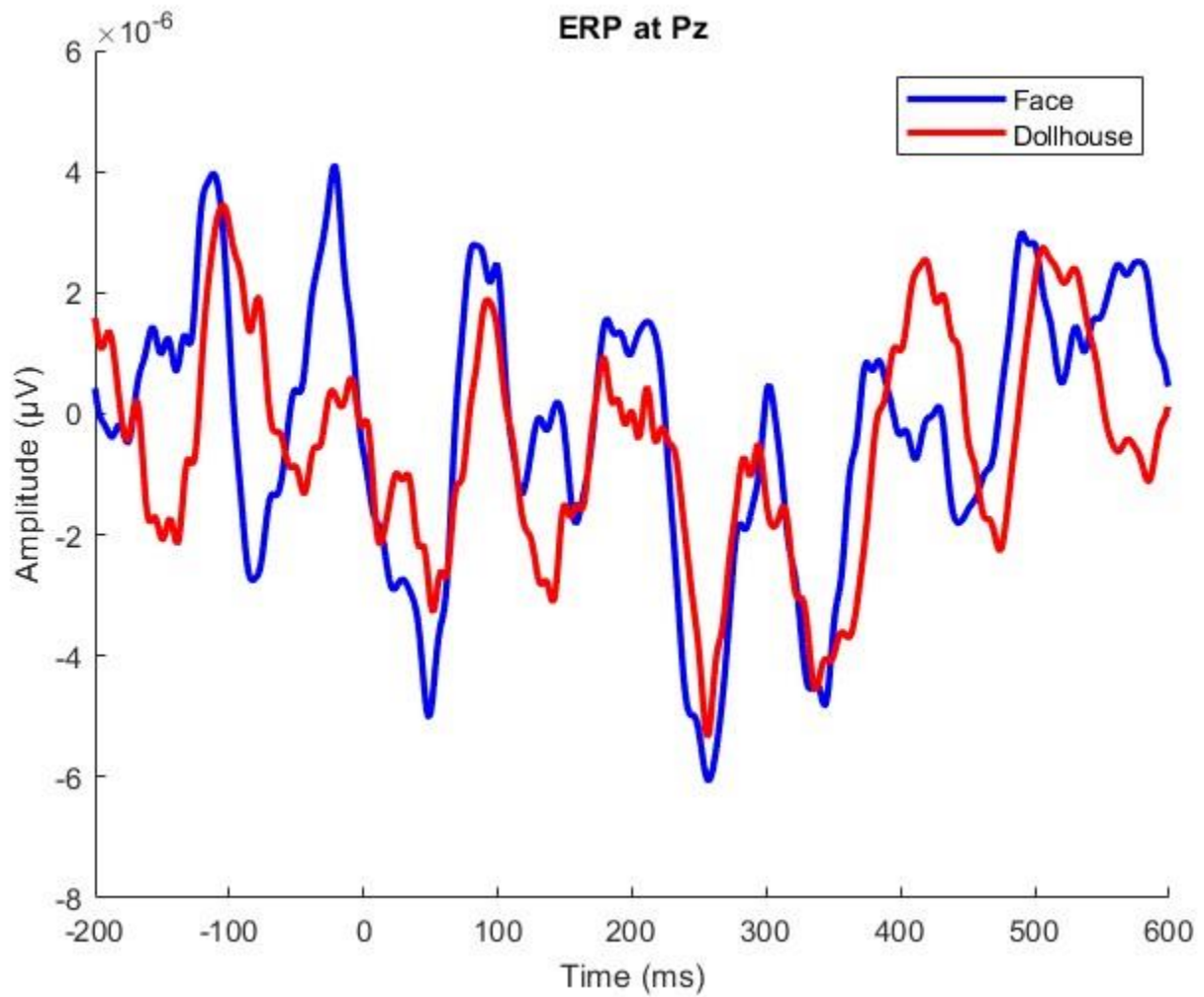


Figure 23 - ERP at Pz channel of both categories

These results suggest that face processing engaged both low-frequency oscillations, potentially linked to attention and memory encoding, and higher-frequency dynamics associated with perceptual integration. Dollhouse stimuli, in contrast, produced more localized beta–gamma bursts.



## Face – Doll difference at 170 ms

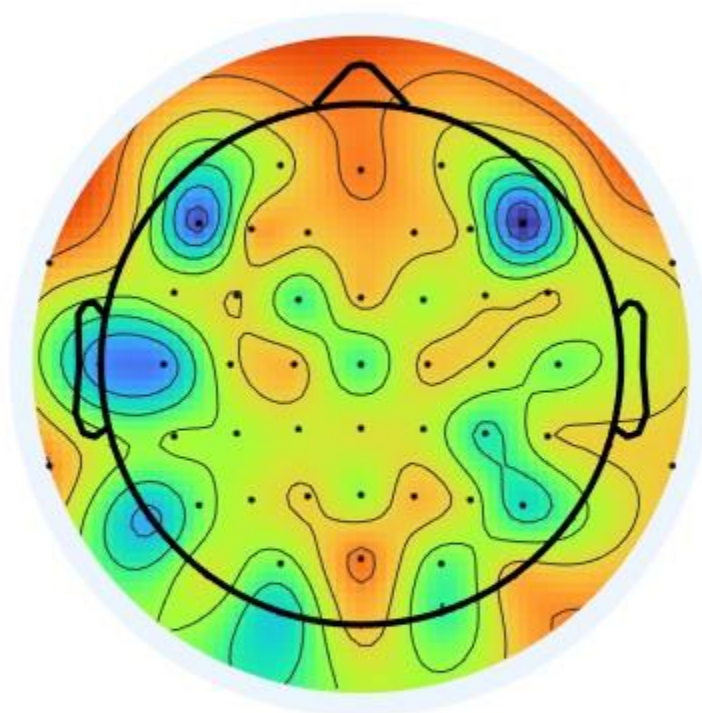


Figure 24 - Face and Dollhouse category difference on scalp in 170 ms

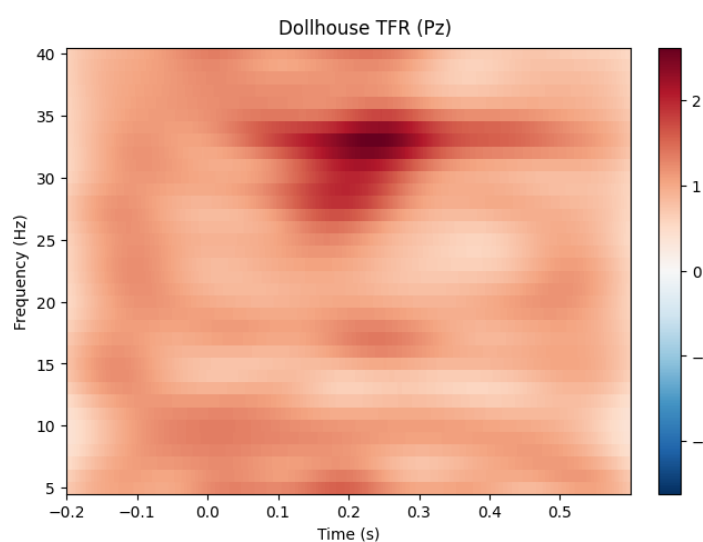


Figure 25 - TFR for dollhouse category

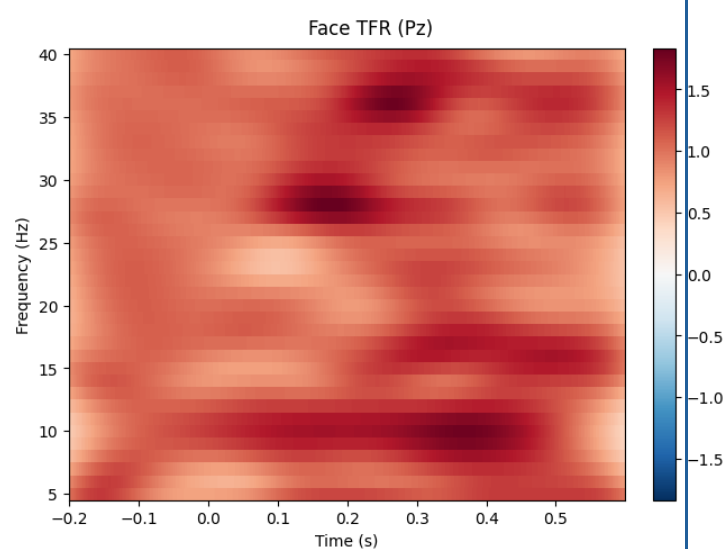
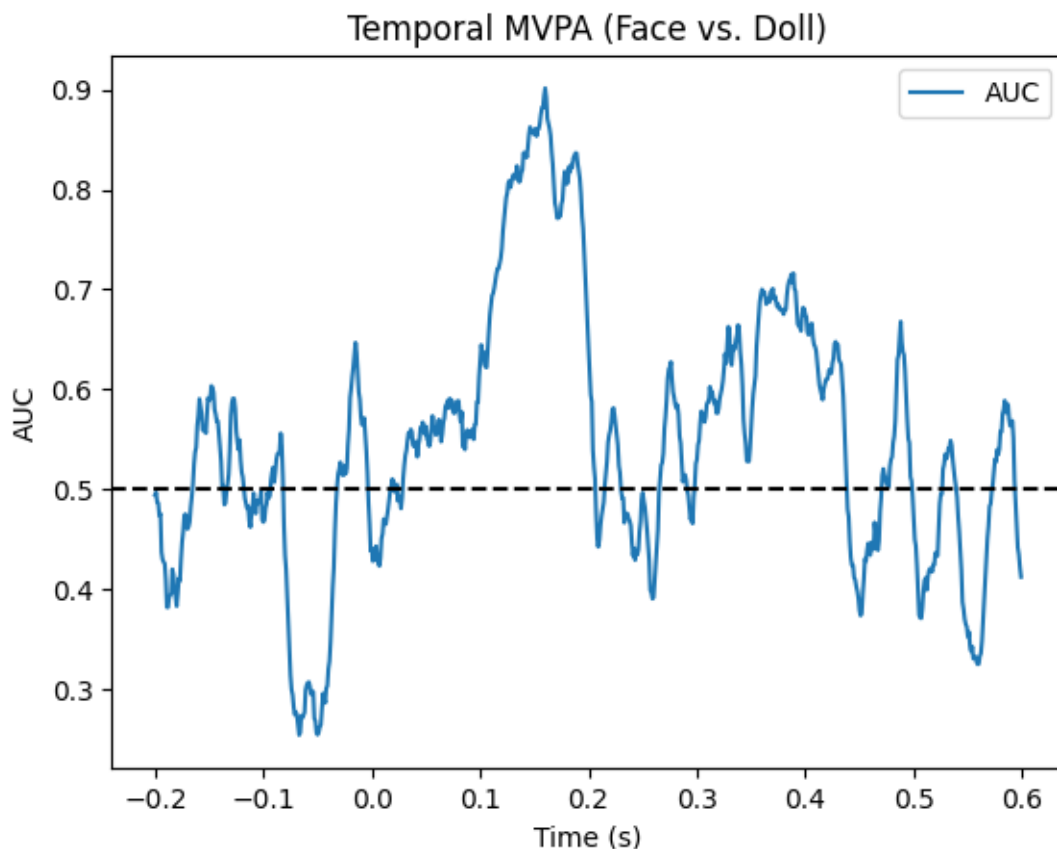


Figure 26 - TFR for face category

## Question 4 – Multivariate pattern analysis (MVPA)

We applied both temporal and spatial MVPA to assess the discriminability of face versus dollhouse trials.

**Temporal MVPA:** Classification accuracy (AUC) rose above chance ( $\sim 0.5$ ) at  $\sim 100$  ms post-stimulus, peaked at  $\sim 0.9$  around 170–200 ms, and remained above chance until  $\sim 400$  ms. This indicates robust temporal dynamics, with maximal separability during the N170 window associated with face-sensitive processing.



*Figure 27 - Temporal MVPA for face and dollhouse categories*

**Spatial MVPA:** When considering the scalp topography of activity patterns in the 170–250 ms window, spatial decoding reached an **AUC of 0.674**, significantly above chance. This demonstrates that not only the temporal evolution but also the **spatial distribution of EEG signals** across electrodes carries information sufficient to discriminate between categories. Notably, this effect was strongest at occipito-temporal electrodes, consistent with the known scalp distribution of face-sensitive activity.



## Question 5 – Representational Dissimilarity Matrix (RDM)

RDMs were constructed from EEG activation patterns across the 20 stimuli (10 faces, 10 dollhouses) at each time slice. These were compared to RDMs derived from CORnet-S layers (V1, V2, V4, IT) using Representational Similarity Analysis (RSA).

Correlations showed that **early EEG time windows (<150 ms)** aligned more closely with **V1/V2 layers**, consistent with early sensory encoding of visual features.

In contrast, **later windows (150–250 ms)** demonstrated higher correspondence with **V4 and IT layers**, paralleling the transition toward categorical object and face representations in the ventral stream.

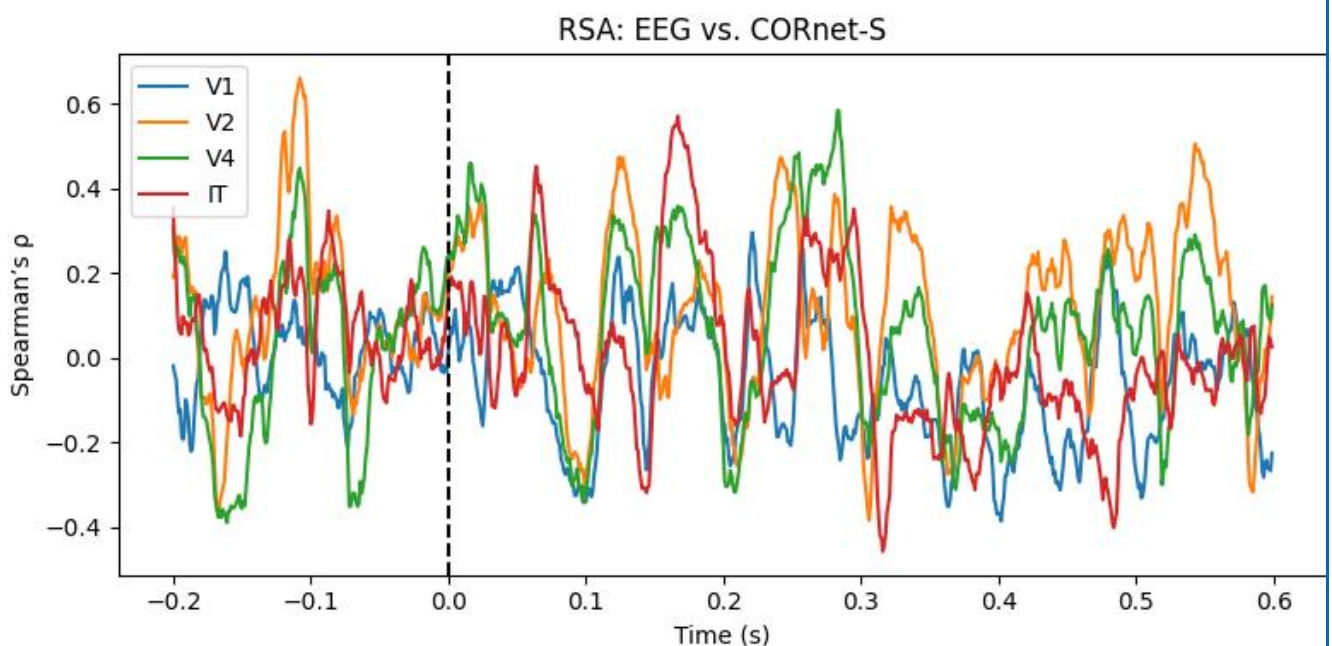


Figure 28 - RSA of different correlations of V1 V2 V4 IT over time

This temporal alignment supports the view that CORnet-S captures aspects of hierarchical ventral stream processing and that EEG can reveal the unfolding of these representations over time.